

13/8/25

Task-2 : Implement conditional, control and looping statements

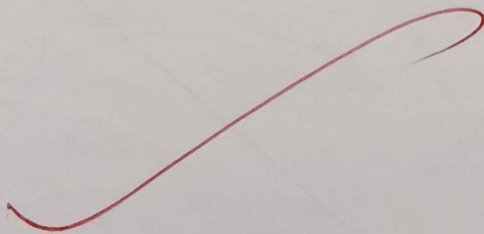
2.1 - You are developing a simple grade management system for a school. The system needs to determine the grade of a student based on their score in a test.

Aim:

To implement conditional, control and looping statements using Python.

Algorithm:

1. Start
2. Get the input mark from the user
3. With the use of an If-elif-else statement do
 - If the marks ≥ 90 print grade "A".
 - If the mark is between 80 and 89 print grade "B".
 - If the mark is between 70 and 79 print grade "C".
 - If the mark is between 60 and 69 print grade "D".
 - If the mark is below 60, print grade "F".
4. Stop



Program:

```
Score = int(input("Enter the score:"))  
if (score >= 90):  
    print("The Grade is A").  
elif (Score <= 89 and score >= 80):  
    print("The Grade is B").  
elif (score <= 79 and score >= 70):  
    print("The Grade is C")  
elif ("score <= 69 and score >= 60):  
    print("The Grade is D")  
else:  
    print("The Grade is F")
```

Output:

Enter the score: 90

The Grade is A

TECH - C25	
1	PERFORMANCE (5)
2	RESULT AND ANALYSIS (3)
3	VIVA VOCE (3)
4	RECORD (4)
5	TOTAL (15)
6	WITH DATE

2.2. The electronics maintenance team at a data center needs a tool to assess the health status of UPS backup batteries based on their current charge percentage.

Aim:

To develop a python program that accepts the battery percentage as input and categorizes the battery health

Algorithm:

1. Accept battery percentage from the user
2. Use ladderized if-elif-else to determine the health category:
 - If percentage $\geq 90 \rightarrow$ "Excellent battery Health"
 - If $70 \leq \text{percentage} < 90 \rightarrow$ "Good battery Health"
 - If $40 \leq \text{percentage} < 70 \rightarrow$ "Average Battery Health"
 - If percentage $< 40 \rightarrow$ "Poor Battery Health"

Python Program:

```
percentage = int(input("Enter battery percentage"))
```

```
if percentage >= 90:
```

```
    print("Excellent Battery health")
```

```
elif percentage >= 70:
```

```
    print("Good Battery Health")
```

```
elif percentage >= 40:
```

```
    print("Average Battery Health")
```

```
else:
```

```
    print("Poor battery Health")
```

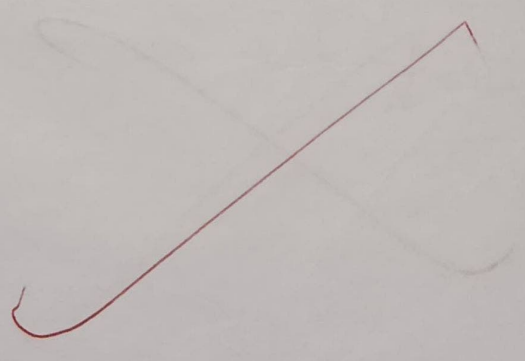
Input:

Battery charge percentage (integer)

Sample Output:

Enter battery percentage: 85

Good Battery Health



2.3. You're coding a system at an amusement park that checks the height of each visitor

Aim:

To implement a python program to allowed each visitor and checks the height of each visitor

Algorithm:

1. Start the program
2. Set the total number of visitors to 5
3. Loop from visitor 1 to visitor 5:
 - Accept the height of the visitor as input (in cm)
 - If height is greater than or equal to 120, print "Allowed"
 - Else print "Not allowed"
4. End the loop after 5 visitors have been checked
5. Stop the program.

VEL TECH	
EX NO.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	15

Result:

Thus, the python program was successfully implemented using conditional statements (if-else), control flow and looping statements.

Program:

```
for i in range(1,6):
```

```
    height = int(input("Enter height of visitor {i} in cm"))
```

```
    if height >= 120:
```

```
        print("Allowed to ride")
```

```
    else:
```

```
        print("Not allowed to ride")
```

Sample Input:

Enter height of visitor 1 in cm : 130

Enter height of visitor 2 in cm : 110

Enter height of visitor 3 in cm : 150

Enter height of visitor 4 in cm : 90

Enter height of visitor 5 in cm : 125

Sample Output:

Allowed

Not Allowed

Allowed

Not Allowed

Allowed

